

3	D	Gradient of displacement time graph is velocity, hence option A, B and C are incorrect. Since displacement decreases after 2.5s, option D is true.
4	C	Net force on trailer = $600 - 200 = 400 \text{ N}$ Hence acceleration on trailer = $400/400 = 1 \text{ m s}^{-2}$ Net force on car = $ma = 1200 \times 1 = 1200 \text{ N}$
5	C	Relative speed of approach: $u_1 - u_2 = v_2 - v_1 = 6 - (-1) = 7$. Only B and C has $v_2 - v_1 = 7$